pashto college girls dance new pashto dance

Question 1

Let the operators $U : H \to H$ and $V : H \to H$ be unitary; here, H is a Hilbert space. Then show that U is normal.

Prove:

$$UU^* = U^*U$$

Since we know that

$$U^* = U^{-1}$$

So

$$UU^{-1} = U^{-1}U$$

$(UV)^* = (UV)^{-1} = I$ Hence its proved that U is normal.

Question 2

Let the operators $U : H \to H$ and $V : H \to H$ be unitary; here, H is a Hilbert space. Then show that $\|U\| = 1$, provided $H \neq \{0\}$.

OR

Let the operators $U : H \to H$ and $V : H \to H$ be unitary; here, H is a Hilbert space. Then show that $U^{-1} (= U^*)$ is unitary.

Prove:

Since U is bijective it shows that U^{-1} is bijective

$$(U^{-1})^* = (U^*)^* = U = (U^{-1})^{-1}.$$

Hence it implies that U^{-1} is unitary.

Question 3

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that U is isometric; thus $\|Ux\| = \|x\|$ for all $x \in H$.

Prove:

$$\| U_x \|^2 = < U_x, U_x >$$
$$\| U_x \|^2 = < x, U^* U_x >$$
$$\| U_x \|^2 = < x, Ix >$$
$$\| U_x \|^2 = < x, x >$$
$$\| U_x \|^2 = \| x \|^2$$

It proved.

Question 4

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $(ST)^* = T^*S^*$ (assuming $H_2 = H_1$) .

Prove:

$$< x, (ST)^* y >=< (ST)x, y >$$

$$< x, (ST)^* y \geq< (SxTx), y >$$

$$< x, (ST)^* y >=< Sx, y >< Tx, y >$$

$$< x, (ST)^* y >=< x, S^* y >< x, T^* y >$$

$$< x, (ST)^* y >=< x, (S^* + T^* y) >$$

$$< x, (ST)^* y >=< x, (S^* T^*) y >$$

Hence we have proved that

$$(ST)^* = S^* T^*$$

Question 5

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $(\alpha T)^* = \bar{\alpha} T^*$.

Prove:

$$< x, (\alpha T^*)^* y > = < (\alpha T)x, y >$$
$$< x, (\alpha T^*)^* y > = < \alpha(Tx), y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < Tx, y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < x, T^* y >$$
$$< x, (\alpha T^*)^* y > = < x, \bar{\alpha} T^* y >$$
$$(\alpha T^*)^* = \bar{\alpha} T^*$$

Question 6

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $(T^*)^* = T$.

Prove:

$$< x, (T^*)^* y \geq < xT^*, y >$$

$$< x, (T^*)^* y \geq < x, Ty >$$

It implies that

$(T^*)^* = T$ so its proved.

Question 7

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $\|T^*T\| = \|TT^*\| = \|T\|^2$.

Prove:

$$\|Tx\|^2 = \langle Tx, Tx\rangle = \langle T^*Tx, x\rangle \leqq \|T^*Tx\|\,\|x\| \leqq \|T^*T\|\,\|x\|^2.$$

Taking the supremum over all x of norm 1, we obtain $\|T\|^2 \leqq \|T^*T\|$.

we have

$$\|T\|^2 \leqq \|T^*T\| \leqq \|T^*\|\,\|T\| = \|T\|^2.$$

Hence $\|T^*T\| = \|T\|^2$. Replacing T by T^* and we also have

$$\|T^{**}T^*\| = \|T^*\|^2 = \|T\|^2.$$

Here $T^{**} = T$

Question 8

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $\langle T^* y, x \rangle = \langle y, Tx \rangle \quad (x \in H_1, y \in H_2)$.

Prove:

$$< T^* y, x > = < \overline{y, T^* x} >$$
$$< T^* y, x > = < \overline{Tx, y} >$$
$$< T^* y, x > = < y, Tx >$$

Hence its proved.

Question 9

Let X and Y be inner product spaces and $Q: X \to Y$ a bounded linear operator. Then Show that $Q = 0$ if and only if $\langle Qx, y \rangle = 0$ for all $x \in X$ and $y \in Y$.

Prove:

$$Q = 0 \Longrightarrow Qx = 0 \; for \; all \; x \epsilon X$$
$$< Qx, y > = < 0, y >$$
$$< Qx, y > = < 0.w, y >$$
$$< Qx, y > = \; 0 < w, y >$$
$$< Qx, y > = \; 0. \quad for \; all \; x \epsilon X$$

It implies that Qx=0 and by definition Q=0. Its proved.

Question 10

Let (T_n) be a sequence of bounded self-adjoint linear operators $T_n : H \to H$ on a Hilbert space H. Suppose that (T_n) converges. Say, $T_n \to T$, that, $\|T_n \to T\| \to 0$, where $\|\cdot\|$ is the norm on the space $B(H,H)$; then show that the limit operator T is a bounded self-adjoint linear operator on H.

Prove:

We show that $T^* = T$. This follows from $\|T - T^*\| = 0$.

$$\|T_n^* - T^*\| = \|(T_n - T)^*\| = \|T_n - T\|$$

and obtain by the triangle inequality in $B(H, H)$

$$\begin{aligned}\|T - T^*\| &\leqq \|T - T_n\| + \|T_n - T_n^*\| + \|T_n^* - T^*\| \\ &= \|T - T_n\| + 0 + \|T_n - T\| \\ &= 2\|T_n - T\| \longrightarrow 0 \quad (n \longrightarrow \infty).\end{aligned}$$

Hence $\|T - T^*\| = 0$ and $T^* = T$.

Question 11

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that a bounded linear operator T on a complex Hilbert space H is unitary if and only if T is isometric and surjective.

Prove:

Suppose that T is isometric and surjective. Isometry implies injectivity, so that T is bijective. We show that $T^* = T^{-1}$. By the isometry,

$$\langle T^*Tx, x\rangle = \langle Tx, Tx\rangle = \langle x, x\rangle = \langle Ix, x\rangle.$$

Hence

$$\langle (T^*T - I)x, x\rangle = 0$$

and $T^*T - I = 0$ so that $T^*T = I$. From this,

$$TT^* = TT^*(TT^{-1}) = T(T^*T)T^{-1} = TIT^{-1} = I.$$

Together, $T^*T = TT^* = I$. Hence $T^* = T^{-1}$, so that T is unitary. The converse is clear since T is isometric and surjective by

Question 12

Let $T : H \to H$ be bounded linear operator on a Hilbert space H . Then show that if H is complex and $\langle Tx, x \rangle$ is real for all $x \in H$, the operator T is self-adjoint.

Prove:

If $\langle Tx, x \rangle$ is real for all x, then

$$\langle Tx, x \rangle = \overline{\langle Tx, x \rangle} = \overline{\langle x, T^*x \rangle} = \langle T^*x, x \rangle.$$

Hence

$$0 = \langle Tx, x \rangle - \langle T^*x, x \rangle = \langle (T - T^*)x, x \rangle$$

and $T - T^* = 0$

since H is complex.

Question 13

Prove that the product of two bounded self-adjoint linear operators S and T on a Hilbert space H is self-adjoint if and only if the operators commute, $ST = TS$.

Prove:

$$< x, (ST)^* y >=< (ST)x, y >$$
$$< x, (ST)^* y \geq< (SxTx), y >$$
$$< x, (ST)^* y >=< Sx, y >< Tx, y >$$
$$< x, (ST)^* y >=< x, S^* y >< x, T^* y >$$
$$< x, (ST)^* y >=< x, (S^* + T^* y) >$$
$$< x, (ST)^* y >=< x, (S^* T^*) y >$$

Hence we have proved that

$$(ST)^* = S^* T^*$$

$$(ST)^* = T^* S^* = TS.$$

Hence

$$ST = (ST)^* \iff ST = TS.$$

Question 14

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $T^*T = 0 \Leftrightarrow T = 0$.

Prove:

$$T^*T = 0 \; iff \; T = 0$$

$$e = \|T^*T\| = \|T\|$$

(f) is true by e.

Queestion 1



Show that $(\alpha T)^* = \alpha^{-1} T^*$

Solution:

$$< x, (\alpha T^*)^* y > = < (\alpha T)x, y >$$
$$< x, (\alpha T^*)^* y > = < \alpha(Tx), y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < Tx, y >$$
$$< x, (\alpha T^*)^* y > = \bar{\alpha} < x, T^* y >$$
$$< x, (\alpha T^*)^* y > = < x, \bar{\alpha} T^* y >$$
$$(\alpha T^*)^* = \bar{\alpha} T^*$$

Education Information Expert
Prepared by Mubashar Ali

Question 2

Prove that $(S+T)^* = S^* + T^*$

Prove:

$$< x, (S+T)^* y > = < (S+T)x, y >$$
$$< x, (S+T)^* y \geq < (Sx+Tx), y >$$
$$< x, (S+T)^* y > = < Sx, y > + < Tx, y >$$
$$< x, (S+T)^* y > = < x, S^* y > + < x, T^* y >$$
$$< x, (S+T)^* y > = < x, (S^* y + T^* y) >$$
$$< x, (S+T)^* y > = < x, (S^* + T^*) y >$$

Hence we have proved that

$$(S+T)^* = S^* + T^*$$

Education promotion Expert
Prepared by Nabeela Ali

Question 3

Find matrix multiplication if $(\xi_1, \xi_2, \xi_3) \leftrightarrow (\xi_1, \xi_2, -\xi_1 - \xi_2)$

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ -\xi_1, -\xi_2 \end{bmatrix}$$

Hence required matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$

Education Information Expert
Prepared by Mubashar Ali

Question 4

$$\|x_1 + x_2 + \ldots\ldots + x_n\|^2 = \|x_1\|^2 + \|x_2\|^2 + \ldots\ldots + \|x_n\|^2$$

Prove:

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \langle \left\|\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i\right\| \rangle$$

$$= < x_1 + \ldots\ldots + x_n, x_1 + \ldots\ldots + x_n >$$

$$= < x_1, x_1 + \ldots\ldots + x_n > + \ldots + < x_n, x_1 + \ldots\ldots + x_n >$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} < x_i, x_j >$$

$< x_i, x_j >= \|x_i\|^2$, if $i \neq j$ $< x_i, x_j >= 0$ and for $i = j$ then $< x_i, x_j >= \|x_i\|^2$

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \sum_{i=1}^{n} \|x_i\|^2$$

Education Information Expert
Prepared by Mudasshar Ali

Question 5

Prove that $(ST)^* = S^*T^*$

Prove:

$$< x, (ST)^*y >=< (ST)x, y >$$
$$< x, (ST)^*y \geq< (SxTx), y >$$
$$< x, (ST)^*y >=< Sx, y >< Tx, y >$$
$$< x, (ST)^*y >=< x, S^*y >< x, T^*y >$$
$$< x, (ST)^*y >=< x, (S^*+T^*y) >$$
$$< x, (ST)^*y >=< x, (S^*T^*)y >$$

Hence we have proved that

$$(ST)^* = S^*T^*$$

Question 6

Show that $\overline{N(T)} < N(T)$

Prove:

$$x \in N(T)$$
$$\exists\ (x_n) such\ that\ x_n\ \in N(T) + x_n \to x$$

Therefore

$$Tx_n \to Tx$$
$$x_n\ \in N(T)$$
$$Tx_n = 0$$
$$T_n = 0$$
$$x\ \in N(T)$$
$$\overline{N(T)} < N(T)$$
$$\overline{N(T)} = N(T)$$
$$N(T) is\ closed.$$

Education Information Expert
Prepared by Mubashir Ali

Question 7

Prove that $< x, ay + bz >= \bar{a} < x, y > + \bar{b} < x, z >$

Prove:

$$< x, ay + bz > = < \overline{ay + bz, x} >$$
$$< x, ay + bz > = < \overline{ay, x} > + < \overline{bz, x} >$$
$$< x, ay + bz > = \bar{a} < \overline{y, x} > + \bar{b} < \overline{z, x} >$$
$$< x, ay + bz > = \bar{a} < x, y > + \bar{b} < x, z >$$

Hence proved.

Education Information Expert
Prepared by Mubashar Ali

Question 8



Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $(T^*)^* = T$.

Prove:

$$< x, (T^*)^* y \geq < xT^*, y >$$

$$< x, (T^*)^* y \geq < x, Ty >$$

It implies that

$(T^*)^* = T$ so its proved.

Education & Information Expert
Prepared by Mr. bashar Ali

Question 9

Prove that the product of two bounded self-adjoint linear operators S and T on a Hilbert space H is self-adjoint if and only if the operators commute, $ST = TS$.

Prove:

$$< x, (ST)^* y >=< (ST)x, y >$$
$$< x, (ST)^* y \geq< (SxTx), y >$$
$$< x, (ST)^* y >=< Sx, y >< Tx, y >$$
$$< x, (ST)^* y >=< x, S^* y >< x, T^* y >$$
$$< x, (ST)^* y >=< x, (S^* + T^* y) >$$
$$< x, (ST)^* y >=< x, (S^* T^*) y >$$

Hence we have proved that

$$(ST)^* = S^* T^*$$

$$(ST)^* = T^* S^* = TS.$$

Hence

$$ST = (ST)^* \iff ST = TS.$$

Question 10

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $\|T^*T\| = \|TT^*\| = \|T\|^2$.

Prove:

$$\|Tx\|^2 = \langle Tx, Tx\rangle = \langle T^*Tx, x\rangle \leqq \|T^*Tx\|\,\|x\| \leqq \|T^*T\|\,\|x\|^2.$$

Taking the supremum over all x of norm 1, we obtain $\|T\|^2 \leqq \|T^*T\|$.

we have

$$\|T\|^2 \leqq \|T^*T\| \leqq \|T^*\|\,\|T\| = \|T\|^2.$$

Hence $\|T^*T\| = \|T\|^2$. Replacing T by T^* and we also have

$$\|T^{**}T^*\| = \|T^*\|^2 = \|T\|^2.$$

Here $T^{**} = T$

Education Information Expert
Prepared by Mubashar Ali

Question 11

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that U is isometric; thus $\|Ux\| = \|x\|$ for all $x \in H$.

Prove:

$$\| U_x \|^2 = < U_x, U_x >$$
$$\| U_x \|^2 = < x, U^* U_x >$$
$$\| U_x \|^2 = < x, Ix >$$
$$\| U_x \|^2 = < x, x >$$
$$\| U_x \|^2 = \| x \|^2$$

It proved.

Education Information Expert
Prepared By Mubashar Ali

Question 12

Show that $< T^*y, x > = < y, Tx >$

Prove:

$$< T^*y, x > = \overline{< y, T^*x >}$$
$$< T^*y, x > = \overline{< Tx, y >}$$
$$< T^*y, x > = < y, Tx >$$

Hence its proved.

Education Information Expert
Prepared by Mubashar Ali

Question 13

Let X and Y be inner product spaces and $Q: X \to Y$ a bounded linear operator. Then Show that $Q = 0$ if and only if $\langle Qx, y \rangle = 0$ for all $x \in X$ and $y \in Y$.

Prove:

$$Q = 0 \Longrightarrow Qx = 0 \; for \; all \; x \epsilon X$$
$$< Qx, y > = < 0, y >$$
$$< Qx, y > = < 0.w, y >$$
$$< Qx, y > = \; 0 < w, y >$$
$$< Qx, y > = \; 0. \quad for \; all \; x \epsilon X$$

It implies that Qx=0 and by definition Q=0. Its proved.

Education Information Experts
Prepared by Mudassar Ali

Question 14

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that $\|U\| = 1$, provided $H \neq \{0\}$.

OR

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that $U^{-1} (= U^*)$ is unitary.

Prove:

Since U is bijective it shows that U^{-1} is bijective

$$(U^{-1})^* = (U^*)^* = U = (U^{-1})^{-1}.$$

Hence it implies that U^{-1} is unitary.

Education Information Expert
Prepared by Mubashir Ali

Question 15

Let the operators $U: H \to H$ and $V: H \to H$ be unitary; here, H is a Hilbert space. Then show that U is normal.

Prove:

$$UU^* = U^*U$$

Since we know that

$$U^* = U^{-1}$$

So

$$UU^{-1} = U^{-1}U$$

$(UV)^* = (UV)^{-1} = I$ Hence its proved that U is normal.

Education Information Expert
Prepared by Mubashar Ali

Question 16

In an inner product space V and x, y in V if $x \perp y$ then

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2$$

Prove:

$$\|x+y\|^2 = < x+y, x+y >$$

$$= < x,x > + < x,y > + < y,x > + < y,y >$$

As x and y are perpendicular so $< x,y >= 0, < y,x >= 0$

$$\|x+y\|^2 = < x,x > + < y,y > = \|x\|^2 + \|y\|^2$$

Education Information Expert
Prepared by Mubashar Ali

Question 17

$$\|x+y\|^2+\|x-y\|^2=2\left(\|x\|^2+\|y\|^2\right) \quad \text{for all x,y } \in V$$

Prove:

$$\|x+y\|^2 = < x+y, x+y >$$
$$= < x,x > + < x,y > + \overline{< x,y >} + < y,y >$$
$$= \|x\|^2 + 2\,\mathrm{Re} < x,y > + \|y\|^2 \quad \text{....(i}$$

Replace y=-y

$$\|x-y\|^2 = < x+y, x+y >$$
$$= < x,x > - < x,y > - \overline{< x,y >} + < y,y >$$
$$= \|x\|^2 - 2\,\mathrm{Re} < x,y > + \|y\|^2 \quad \text{.....(ii}$$

Adding (i and (ii

$$\|x+y\|^2+\|x-y\|^2=2\|x\|^2+2\|y\|^2$$

That we have to prove.

Education Information Expert
Prepared By Mubashar Ali

Question 17

$$\|x+y\|^2+\|x-y\|^2=2\left(\|x\|^2+\|y\|^2\right) \quad \text{for all x,y } \in V$$

Prove:

$$\|x+y\|^2 = < x+y, x+y >$$
$$= < x,x > + < x,y > + \overline{< x,y >} + < y,y >$$
$$= \|x\|^2 + 2\,\mathrm{Re} < x,y > + \|y\|^2 \quad \text{....(i}$$

Replace y=-y

$$\|x-y\|^2 = < x+y, x+y >$$
$$= < x,x > - < x,y > - \overline{< x,y >} + < y,y >$$
$$= \|x\|^2 - 2\,\mathrm{Re} < x,y > + \|y\|^2 \quad \text{.....(ii}$$

Adding (i and (ii

$$\|x+y\|^2+\|x-y\|^2=2\|x\|^2+2\|y\|^2$$

That we have to prove.

Education Information Expert
Prepared By Mubashar Ali

Question18

For any two elements x,y is an inner product space V,

$$|< x, y >| \leq \|x\|.\|y\|$$

Prove:

If x=y=0 then 0=0

Let at least one of x and y is not equal to zero

Let $|< x+\lambda y, x+\lambda y >| \geq 0$ by definition

$$< x, x+\lambda y > + < \lambda y, x+\lambda y >$$

$$< x, x+\lambda y > + y < y, x+\lambda y >$$

Education Information Expert
Prepared by Nusrat Shar Ali

Question 19

SHOW THAT T is uniquely determinal if the image $y_k = Te_k$ of n basis vectors $e_1, \ldots\ldots\ldots, e_n$ are prescribed.

Prove:

$$y=Tx \;; y \in Y \;\; \{b_1, \ldots\ldots\ldots, b_n\}$$

$$y=\eta_1 b_1 + \eta_2 b_2 + \ldots\ldots\ldots + \eta_r b_r$$

$$Te_k \in Y, \qquad Te_1 = \tau_{11} b_1 + \tau_{12} b_2 + \ldots\ldots\ldots + \tau_{1r} b_r$$

$$Te_k = \sum_{j=1}^{r} \tau_{kj} b_r$$

$$y=\sum_{j=1}^{r} \eta_j b_j = \sum_{k=1}^{n} \xi_k Te_k = \sum_{k=1}^{n} \xi_k \sum_{j=1}^{r} \tau_{kj} b_j$$

Comibinig these two summation

$$y = \sum_{j=1}^{r} \left(\sum_{k=1}^{n} \tau_{kj} \xi_k \right) b_j$$

$$\eta_j = \sum_{k=1}^{n} \tau_{kj} \xi_k$$

The image $y=Tx=\sum \eta_j b_j$ of $x = \sum \xi_k Te_k$ can be obtained from

$$\eta_j = \sum_{k=1}^{n} \tau_{kj} \xi_k$$

Education Information Expert
Prepared By Mubashar Ali

Question 20

Sum and Scalar multipication of Dual.

Prove:

as follows. The SUM $f_1 + f_2$ of two **functionals** f_1 **and** f_2 is the functional s whose value **at every** $x \in X$ is

$$s(x) = (f_1 + f_2)(x) = f_1(x) + f_2(x);$$

the Product αf of a scalar α and **a functional** f is the functional p whose value **at** $x \in X$ is

$$p(x) = (\alpha f)(x) = \alpha f(x).$$

Education Information Expert
Prepared by Mudassar Ali

Question 21

Define Inner Product Space:

The pair $(V, \langle \bullet, \bullet \rangle)$ is called an inner product space.

Define Appolonius Identity:

$$\|z-x\|^2+\|z-y\|^2=\frac{1}{2}\|x-y\|^2+2\left\|z-\frac{1}{2}(x+y)\right\|^2 \quad , \quad x,y,z \in V$$

Define Equal Operators:

Let T_1 and T_2 be two operators defined on equal i.e. $T_1 = T_2$ if they have the same domain i.e. $D(T_1) = D(T_2)$

Question 22

Define Orthonormal System.

Orthonormal systems are those functions which are real or complex whose inner product with itself results in 1 and with other functions result in 0.

Define Polarization Identity.

For any x, y in complex inner product space

$$< x, y >= \frac{1}{4}\left\{\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2\right\}$$

We have to prove this complex inner product space.

Question 23

Define Bounded Linear Function.

Let X and Y be normed spaces and $T : D(T) \to Y$ a linear operator, where $D(T) \subset X$. The operator T is said to be bounded if there is a real number c such that for all $x \in D(T)$.

$\|Tx\| \leq c\|x\|$

Define Canonical Mapping.

$C : X \to X^{**}$ this mapping is called canonical mapping of X into X^{**} defined as $x \mapsto g_x$.

Question 24

Define Unitary Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $T^* = T^{-1}$.

Define Normal Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $TT^* = T^*T$

Define Extension of T.

Extension of T on a set $M \supset D(T)$ is defined as $\tilde{T}: M \to Y$ or

$$\tilde{T}x = Tx \; for \; all \; x \in D(T)$$

Question 25

Write examples of Bounded Linear Operators

Identity Operator, Zero Operator, Differentiation Operator, Integral Operator

Define Self Adjoint Operator or Hermitian Operator.

Abounded Linear Operator $T: H \to H$ on a Hilbert Space H is said to be Self Adjoint Operator or Hermitian Operator if $T^* = T$.

Give an example of function which is not linear.

$\| . \|: x \to R$ is functional but not linear.

Find basis of R(T).

$(\xi_1, \xi_2, -\xi_1, -\xi_2) = \xi_1(1,0,-1) + \xi_2(0,1,-1)$

So the basis of R(T)= $\{(1,0,-1),(0,1,-1)\}$

Question 26

Give any two examples of Linear Functional.

Space C[a, b] and Space l^2

Define Linear Functional:

A linear functional f is a linear operator with domain in a vector space X and range in the scalar field K of X; thus $f: D(f) \to K$. where K=R if X is real and K=C if X is complex.

Define Bounded Linear Extension:

$T: D(T) \to Y$ be a bounded linear operator, where D(T) lies in a normal space X and Y is a Banach Space then T has an extension $\bar{T}: \overline{D(T)} \to Y$. Where $\bar{T}$ is a bounded linear operator of norm.

$$\| \bar{T} \| = \| T \|$$

Education Information Expert
Prepared by Mubashar Ali